INCH - POUND

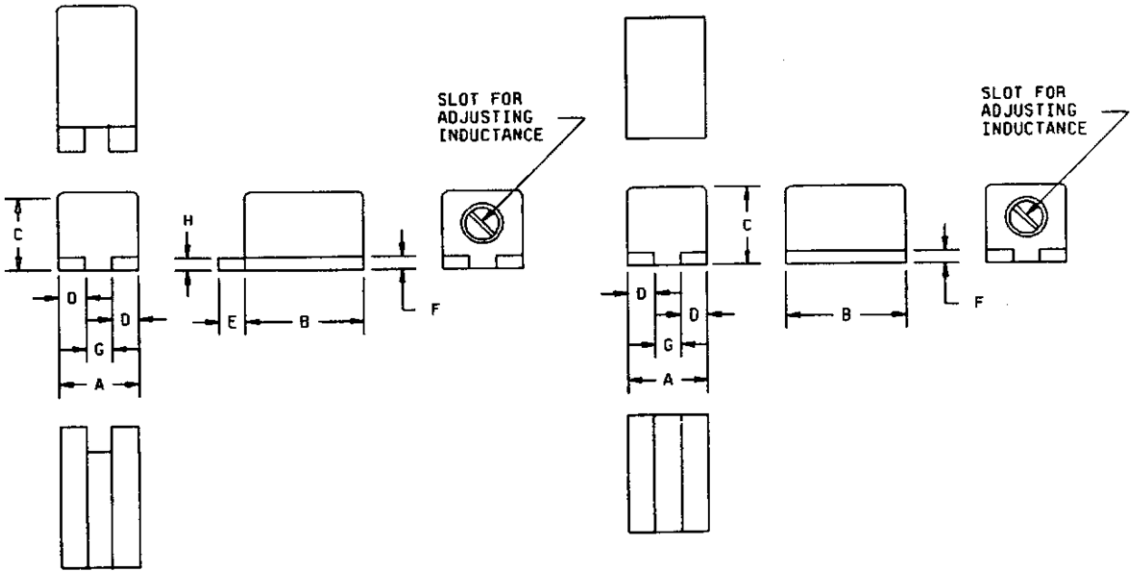
MIL-PRF-83446/7D
w/AMENDMENT 2
28 May 2024
SUPERSEDING
MIL-PRF-83446/7D
w/AMENDMENT 1
3 December 2018

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, VARIABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-83446.



For dash numbers 01 through 47 (with tabs)

For dash numbers 48 through 94 (without tabs)

FIGURE 1. Variable, chip coil.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5950

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For dash numbers 01 through 47 (with tabs)		
Ltr	Inches	mm
A	.115 max	2.92 max
B	.170 max	4.32 max
C	.155 max	2.92 max
D	.030 ±.005	0.76 ±0.13
E	.040 +.015 -.010	1.02 +0.38 -0.25
F	.007 ±.003	0.18 ±0.08
G	.038 ±.010	0.97 ±0.25
H	.005 ±.002	0.13 ±0.05

For dash numbers 48 through 94 (without tabs)		
Ltr	Inches	mm
A	.115 max	2.92 max
B	.170 max	4.32 max
C	.155 max	2.92 max
D	.030 ±.005	0.76 ±0.13
F	.007 ±.003	0.18 ±0.08
G	.038 ±.010	0.97 ±0.25

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Variable, chip coil – Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 0.5 gram, maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: MIL-STD-202-301, test voltage 700 volts rms.

Barometric pressure: MIL-STD-202-105, test condition C (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Supersession data: A new part numbering system with codes for termination materials was incorporated with Revision A, superseding MIL-I-83446/7, dated 1 March 1979.

M83446/07-(dash number from table I) supersedes M83446/7-(dash number from table I).

Examples:

M83446/07-21A supersedes M83446/7-21.
M83446/07-03A supersedes M83446/7-3.

Part of Identifying Number (PIN): The part number shall be in the following form.

M83446/07-	04	B	M
	Sequentially assigned dash Numbers (see table I)	Termination finish (see MIL-PRF-83446)	Inductance tolerance (see MIL-PRF-83446)

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TABLE I. Electrical characteristics (initial) and dash numbers.

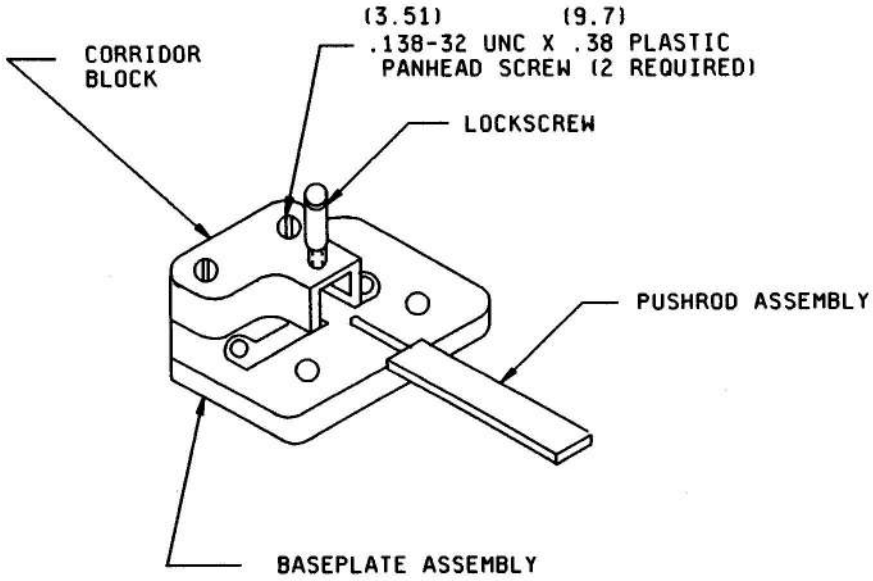
Dash number	Inductance (μH)			Q Min at L nom	Q Typ at L nom	Test Freq-qency (MHz) 1/	Self resonant frequency (min) MHz	DC resistance at 25°C (max) ohms	Current (max) mA 2/	Incre-mental current (max) mA 3/
	Min	Nom	Max							
01, 48	0.016	0.018	0.019	50	60	150	1200	0.02	750	----
02, 49	0.019	0.022	0.024	50	60	150	1100	0.02	750	----
03, 50	0.024	0.027	0.029	50	60	150	1000	0.03	750	----
04, 51	0.029	0.033	0.036	50	59	150	900	0.03	750	----
05, 52	0.036	0.039	0.042	50	60	150	750	0.03	750	----
06, 53	0.042	0.047	0.051	50	59	100	660	0.04	750	----
07, 54	0.051	0.056	0.062	50	60	100	610	0.04	750	----
08, 55	0.061	0.068	0.074	50	58	100	550	0.05	750	----
09, 56	0.072	0.082	0.092	50	60	100	520	0.06	640	----
10, 57	0.091	0.100	0.114	50	58	50	490	0.07	600	----
11, 58	0.112	0.120	0.135	40	50	50	460	0.07	600	----
12, 59	0.130	0.15	0.165	40	50	50	430	0.10	480	----
13, 60	0.160	0.18	0.195	40	50	50	385	0.12	450	----
14, 61	0.195	0.22	0.245	40	51	50	330	0.15	400	----
15, 62	0.240	0.27	0.300	40	50	50	275	0.20	350	----
16, 63	0.295	0.33	0.365	40	50	50	240	0.25	300	----
17, 64	0.360	0.39	0.430	35	46	50	210	0.35	260	----
18, 65	0.425	0.47	0.515	35	42	25	190	0.40	250	----
19, 66	0.510	0.56	0.620	35	43	25	165	0.50	230	----
20, 67	0.610	0.68	0.750	35	42	25	155	0.75	180	----
21, 68	0.740	0.82	0.920	35	41	25	145	0.90	160	----
22, 69	0.900	1.0	1.10	35	42	25	130	1.8	120	----
23, 70	1.10	1.2	1.30	35	43	25	120	1.9	115	----
24, 71	1.35	1.5	1.65	22	33	7.9	105	3.0	90	----
25, 72	1.60	1.8	1.95	22	32	7.9	95	3.5	85	----
26, 73	1.95	2.2	2.45	19	25	7.9	85	6.5	62	----
27, 74	2.40	2.7	3.10	19	26	7.9	80	7.0	60	----
28, 75	3.00	3.3	3.65	15	24	7.9	75	7.5	55	----
29, 76	3.60	3.9	4.40	15	22	7.9	70	10.0	50	----
30, 77	4.30	4.7	5.20	15	26	7.9	38	2.5	100	----
31, 78	5.10	5.6	6.10	20	27	7.9	36	2.5	100	----
32, 79	6.10	6.8	7.35	20	28	7.9	35	3.5	85	----
33, 80	7.3	8.2	9.4	20	28	7.9	30	4.5	75	----
34, 81	9.0	10	11.0	20	28	7.9	25	5.0	70	----
35, 82	10.8	12	13.6	20	28	7.9	20	5.5	65	----
36, 83	13.4	15	16.8	20	30	2.5	18	6.5	60	----
37, 84	16.4	18	20.0	20	30	2.5	17	11.0	47	----
38, 85	19.6	22	25.5	20	30	2.5	16	12.0	45	----
39, 86	24.5	27	31.0	20	30	2.5	13	13.5	43	----
40, 87	30.0	33	38.0	20	30	2.5	11	14.0	42	----
41, 88	36.0	39	44.0	20	30	2.5	10	18.0	37	7.5
42, 89	42.0	47	53.0	20	30	2.5	9	20.0	35	7.0
43, 90	50.0	56	64.0	20	30	2.5	8	22.0	34	6.8
44, 91	60.0	68	76.0	20	30	2.5	7	24.0	32	6.4
45, 92	72.0	82	94.0	20	30	2.5	6	38.0	25	5.0
46, 93	90.0	100	115.0	20	28	2.5	5.2	42.0	24	4.8
47, 94	110.0	120	140.0	20	28	2.5	5.0	52.0	22	4.5

1/ Test frequency range 2.5 through 25 MHz: For electrical characteristics measurements, use the TF-260Q-1 test fixture, or equivalent (see figure 3). Fixture inductance (approximately 0.028 μH) and residual Q-meter inductance (approximately 0.01 μH) should be subtracted from indicated inductance.

Test frequency range 25 through 150 MHz: For electrical characteristics measurements, use the TF-250RX-1 test fixture, or equivalent (see figure 2). Fixture inductance (approximately 0.009 μH) should be subtracted from indicated inductance.

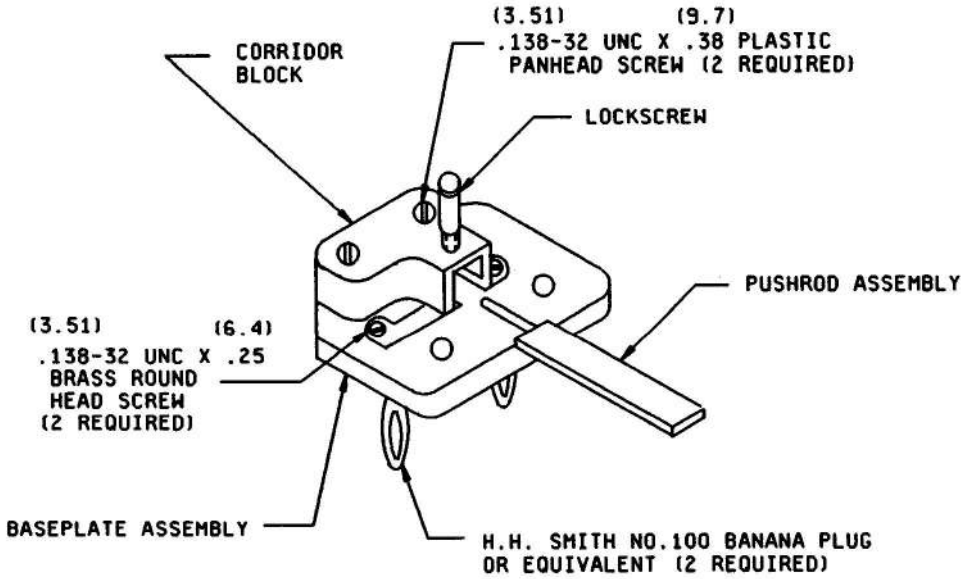
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2/ Maximum current allowed is not to exceed the specified temperature rise.
3/ Applicable to ferrite core only.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Typical chip coil test fixture (TF-250RX-1, or equivalent).



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Typical chip coil test fixture TF-260Q-1, or equivalent).

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurements			
	Inductance (Percent)	DC resistance	Self-resonant frequency (Percent)	Q (Percent)
Qualification inspection				
Group II	±5	±(3% +.001 ohm)	-8	-10
Group IV	±5	±(2% +.001 ohm)	-10	-10
Group V	±2	---	---	-10
Conformance inspection group C				
Subgroup II	±5	±(3% +.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +.001 ohm)	-8	-10

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-83446, this document references the following:

MIL-STD-202-105 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA - CC

Review activities:
Army – MI
Navy – AS, OS
Air Force - 19

Preparing activity:
DLA – CC

(Project 5950-2024-005)

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